

Homework 3

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Problem 1. Using only the fundamental algebraic properties of R , prove the following for $a, b \in \mathbb{R}$:

1. If $a + b = 0$ then $b = -a$
2. $(-1)a = -a$
3. $-(-a) = a$
4. $(-1)(-1) = 1$

Claim 1a. If $a + b = 0$ then $b = -a$

Proof. $\forall a, b \in \mathbb{R}$, suppose $a + b = 0$. By the existence of negative numbers, $\exists -a \in \mathbb{R} \ni a + (-a) = 0$ and $(-a) + a = 0$. Thus, by left adding by $-a$ to both sides,

$$0 + b = -a + a + b = -a + 0$$

.

By the existence of the 0 element and its definition, $0 + b = b$ and $-a + 0 = -a$. Thus $b = -a$. $\square$

Claim 1b. $(-1)a = -a$

Proof. $\forall a \in \mathbb{R}$, by distributivity of multiplication, $(-1)a + a = a(-1 + 1)$. By the existence of negative numbers, $-1 + 1 = 0$. So $a(-1 + 1) = a(0)$.

$\forall r \in \mathbb{R}$, by distributivity of multiplication, $r + r * 0 = r(1 + 0)$. By the existence of zero element and its properties, $1 + 0 = 1$. Thus $r(1 + 0) = r * 1$. By the properties for the unit element, $r * 1 = r$. Thus $r + r * 0 = r$. By the existence of negative numbers, $-r + r = 0$. Thus $-r + r + r * 0 = -r + r$, thus $r * 0 = 0$.

Thus $a * 0 = 0$. Thus $(-1)a + a = 0$. By the existence of negative numbers, $(-1)a + 0 = (-1)a + a + (-a) = 0 + (-a)$. By the properties of 0 element, $(-1)a = -a$. $\square$

Claim 1c. $-(-a) = a$

Proof. $\forall n \in \mathbb{R}$, $n + (-n) = 0$. $\forall a \in \mathbb{R}$, substitute n for a , thus $a + (-a) = 0$. Since addition commutes in $\mathbb{R}$, $(-a) + a = 0$. Substitute n for $-a$ thus $(-a) + (-(-a)) = 0$. Thus $a + (-a) = (-a) + (-(-a))$. Addition commutes in $\mathbb{R}$, thus $a + (-a) = (-(-a)) + (-a) \implies a + (-a) + a = (-(-a)) + (-a) + a$. Thus $a + 0 = (-(-a)) + 0$. By properties of the 0 element in $\mathbb{R}$, $a = a + 0 = (-(-a)) + 0 = (-(-a)) \implies a = -(-a)$. $\square$

Claim 1d. $(-1)(-1) = 1$

Proof. $\forall x \in \mathbb{R}$, by the existence of a multiplicative identity and its properties, $x * 1 = x$. By the existence of an additive identity and its properties, $1 + 0 = 1$. Thus $x * (1 + 0) = x$. By the distributivity of multiplication, $x * 1 + x * 0 = x(1 + 0) = x$. By the properties of the multiplicative identity, $x + x * 0 = x * 1 + x * 0 = x$. By the existence of additive inverse, $\exists -x \in \mathbb{R} \ni x + (-x) = 0$ and $(-x) + x = 0$ since addition is commutative. Thus $(-x) + x + x * 0 = (-x) + x$. Thus $0 + x * 0 = 0 \implies x * 0 = 0$, by the properties of the additive identity.

By the existence of additive inverses, $1 + (-1) = 0$. $-1(1 + (-1)) = -1(0) = 0$, by the above. Since multiplication is distributive, $-1(1) + (-1)(-1) = -1(1 + (-1)) = 0$. By the properties of the unit element of $\mathbb{R}$, $-1(1) = -1$, thus, $-1 + (-1)(-1) = 0$. Add 1 to both sides, since by the existence of additive inverses, $1 + (-1) = -1 + 1 = 0$. Thus $1 + (-1) + (-1)(-1) = 0 + (-1)(-1) = 0 + 1$. By the properties of the additive identity, $(-1)(-1) = 1$. $\square$

Problem 2. Solve the following equations, justifying each step by referring to the fundamental algebraic properties of $\mathbb{R}$ or theorems:

1. $3x + 5 = 10$

2. $x^2 = 3x$

3. $x^2 - 3 = 6$

4. $(x - 3)(x + 5) = 0$

Problem 2a. $3x + 5 = 10$

Proof. By the existence of additive inverse, $\exists -5 \in \mathbb{R} \ni 5 + (-5) = 0$. Add -5 to both sides, thus $3x + 0 = 3x + 5 - 5 = 10 - 5 = 5$. By the properties of the additive identity, $3x + 0 = 3x$. Thus $3x = 5$. By the existence of multiplicative inverses for all numbers not equal to zero, $\exists \frac{1}{3} \in \mathbb{R} \ni 3(\frac{1}{3}) = 1$. Multiply both sides by $\frac{1}{3}$, thus $1x = (\frac{1}{3})3x = 5(\frac{1}{3})$. By the properties of the multiplicative identity, $1x = x$. Thus $x = 5(\frac{1}{3}) = \frac{5}{3}$. $\square$

Problem 2b. $x^2 = 3x$

Proof. Since an additive inverse exists for all real numbers, $\exists -3x \in \mathbb{R} \ni 3x + (-3x) = 0$. Add $-3x$ to both sides, thus $x^2 - 3x = 0$. Since multiplication is distributive in $\mathbb{R}$, $x(x - 3) = x^2 - 3x = 0$. Thus $x(x - 3) = 0$.

$\forall a, b \in \mathbb{R}$, if $ab = 0$. Suppose $a \neq 0$, then by the existence of multiplicative identity for nonzero element $\exists \frac{1}{a} \in \mathbb{R} \ni a(\frac{1}{a}) = 1$. Multiply both sides by $\frac{1}{a}$, thus $\frac{1}{a}ab = 0 * \frac{1}{a}$. Since multiplication is associative and commutative, $\frac{1}{a}ab = (\frac{1}{a} * a) * b = (a * \frac{1}{a}) * b = 1 * b$. By the properties of the multiplicative identity, $1 * b = b$. Since 0 multiplied by any real number is 0 as proven in 1d, $0 * \frac{1}{a} = 0$. Thus $b = 0$.

Thus either $x = 0$ or $x - 3 = 0$. Thus $x = 0$ is a solution. If $x - 3 = 0$, by the properties of an additive inverse, $-3 + -(-3) = 0$, thus $x - 3 + (-(-3)) = 0 + (-(-3)) \implies x + 0 = 0 + (-(-3))$. By the properties of 0, $x + 0 = x$ and $0 + (-(-3)) = -(-3)$. Thus $x = -(-3)$. By 1c $-(-3) = 3$. Thus $x = 3$. $\square$

Problem 2c. $x^2 - 3 = 6$

Proof. By the existence of additive inverse, $\exists -6 \in \mathbb{R} \ni 6 + (-6) = 0$. Thus $x^2 - 3 = 6 \implies x^2 - 3 - 6 = 6 + (-6) = 0 \implies x^2 - 9 = 0$. By the existence of additive inverses, $3x + (-3x) = 0$. By the properties of the additive identity 0, $0 = x^2 - 9 = x^2 - 9 + 0$. Since additive is commutative, $x^2 + 0 - 9 = 0$. Substitute $3x + (-3x)$ for 0, thus $x^2 + 3x - 3x - 9 = x^2 + 0 - 9 = 0$. Since multiplication is distributive, $x(x + 3) = x^2 + 3x$ and $-3(x + 3) = -3x + (-3)(3)$. By claim 1a, $-3 = -1(3)$. Thus $-3(3) = -1(3)(3) = -1(9)$. Again by 1a, $-1(9) = -9$. Thus $-3(x + 3) = -3x + (-3)(3) = -3x - 9$. Thus $x(x + 3) - 3(x + 3) = x^2 + 3x - 3x - 9$. Since multiplication is distributive and commutative, $(x + 3)(x - 3) = x(x + 3) - 3(x + 3)$. Thus $(x + 3)(x - 3) = x^2 - 9 = 0$. By problem 2b, if the product of 2 numbers is 0 either one of the numbers is 0. Thus $x + 3 = 0$ or $x - 3 = 0$. If $x + 3 = 0$. By the properties of additive inverse $3 + (-3) = 0$ and by the properties of additive identity, $x = x + 0 = x + 3 - 3 = 0 + (-3) = -3$. Thus $x = -3$. If $x - 3 = 0$, by the properties of additive inverse $-3 + (-(-3)) = 0$, and by the properties of additive identity, $x = x + 0 = x - 3 + (-(-3)) = 0 + (-(-3)) = (-(-3))$. By 1c, $(-(-3)) = 3$. Thus $x = 3$. Thus $x = -3$ and $x = 3$. $\square$

Problem 2d. $(x - 3)(x + 5) = 0$

Proof. By problem 2b, if the product of 2 numbers is 0 either one of the numbers is 0. Thus either $x - 3 = 0$ or $x + 5 = 0$. Suppose $x - 3 = 0$. By the existence of additive inverses, $\exists -(-3) \in \mathbb{R} \ni -3 + (-(-3)) = 0$. Thus $x + 0 = x - 3 + (-(-3)) = 0 + (-(-3))$. By properties of 0 and by 1c, $x = (-(-3)) = 3$. Suppose $x + 5 = 0$. By the existence of additive inverses, $\exists -5 \in \mathbb{R} \ni 5 + (-5) = 0$. Thus $x + 0 = x + 5 - 5 = 0 - 5$. By properties of 0, $x = x + 0 = 0 + (-5) = -5$. Thus $x = 3$ and $x = -5$. $\square$

Problem 3. Prove $\forall x, y \in \mathbb{R} \ni x \neq 0, y \neq 0, \frac{1}{xy} = \frac{1}{x} * \frac{1}{y}$

Proof. By properties of multiplicative inverse, $\frac{1}{xy} * xy = 1$. Multiply both sides by $\frac{1}{x} * \frac{1}{y}$. Thus $\frac{1}{xy} * xy * \frac{1}{x} * \frac{1}{y} = 1 * \frac{1}{x} * \frac{1}{y}$. By properties of the additive identity and the fact that multiplication is commutative, $\frac{1}{xy} * x * \frac{1}{x} * y * \frac{1}{y} = \frac{1}{x} * \frac{1}{y}$. By the associativity of multiplication and the properties of the multiplicative inverses, $\frac{1}{xy} * 1 * 1 = \frac{1}{xy} * (x * \frac{1}{x}) * (y * \frac{1}{y}) = \frac{1}{x} * \frac{1}{y}$. By the properties of the multiplicative identity 1, $\frac{1}{xy} * 1 * 1 = \frac{1}{xy}$. Thus $\frac{1}{xy} = \frac{1}{x} * \frac{1}{y}$. $\square$

Problem 4. If $y \in R$ and $yy = y$, prove that either $y = 0$ or $y = 1$.

Proof. If $y \in \mathbb{R}$, $y * y = y$. Then by the existence of a additive inverse, $y + (-y) = 0$, thus $y^2 - y = y * y - y = y - y = 0$. Due to multiplication being distributive, $y(y - 1) = y^2 - y = 0$. By problem 2b, $\forall a, b \in \mathbb{R}$ if $ab = 0 \implies a = 0$ or $b = 0$. Thus either $y = 0$ or $y - 1 = 0$. By the existence of a additive inverse for any real, $\exists -(-1) \in \mathbb{R} \ni 1 + (-(-1)) = 0$. Thus $y - 0 = y - 1 + (-(-1)) = 0 - (-(-1))$. By properties of the additive identity, $y = -(-1)$. By 1c, $y = 1$. Thus $y = 0$ or $y = 1$. $\square$

Problem 5. Let $c \in \mathbb{R}$:

1. If $0 < c < 1$ show that $0 < c^2 < c < 1$.
2. If $1 < c$, show that $0 < c < c^2$

Claim 5a. If $0 < c < 1$ show that $0 < c^2 < c < 1$.

Proof. Since $c > 0$ and the fact the multiplication of any positive number doesn't change the direction of the ordering symbol, thus $0c < c^2 < 1c$. By the properties of the unit element, $1c = c$ and by $1d$, $0c = 0$. Thus $0 < c^2 < c$. By assumption, $c < 1$. By the transitive properties of ordering, since $c^2 < c$ and $c < 1$ it means $c^2 < 1$. Thus, $0 < c^2 < c < 1$. $\square$

Claim 5b. Since $1 < c$ and $0 < 1$, by transitive properties of ordering, this means, $0 < c$. Thus c is positive and the multiplication of a positive real on both sides of a inequality doesn't change the direction of the inequality, thus $1c < c^2$. By the properties of the multiplicative identity, $1c = c$. Thus $c < c^2$. Furthermore since $0 < c$ and $c < c^2$, by the transitive properties of order, $0 < c^2$. Putting all the inequalities together, we get, $0 < c < c^2$.

Problem 6. Let $a, b \in \mathbb{R}$:

1. Assume a and b are rational numbers. Prove that $a + b$ and ab are also rational numbers.
2. Prove that if a is a rational number and b is an irrational number, then $a + b$ is an irrational number. If in addition $a \neq 0$, show that ab is an irrational number.

Claim 6a. Assume a and b are rational numbers. Prove that $a + b$ and ab are also rational numbers.

Proof. Suppose $\forall a, b \in \mathbb{Q}$, then by definition of rational numbers, $\exists p, q, m, n \in \mathbb{Z} \ni a = \frac{p}{q}, b = \frac{m}{n}, q \neq 0, n \neq 0$. Thus $a + b = \frac{p}{q} + \frac{m}{n} = \frac{pn}{qn} + \frac{mq}{qn} = \frac{pn+mq}{qn}$. Since $p, n, m, q \in \mathbb{Z}$, $pn, mq, qn \in \mathbb{Z}$, thus $pn + mq \in \mathbb{Z}$. Thus $a + b = \frac{pn+mq}{qn} \in \mathbb{Q}$.

$ab = \frac{pm}{qn}$, as stated above $qn \in \mathbb{Z}$. But $pm \in \mathbb{Z}$ as well. Thus $ab = \frac{pm}{qn} \in \mathbb{Q}$. $\square$

Claim 6b. Prove that if a is a rational number and b is an irrational number, then $a + b$ is an irrational number. If in addition $a \neq 0$, show that ab is an irrational number.

Proof. Suppose $\forall a \in \mathbb{Q}$ and b is any irrational number.

Assume the contrary that $a + b \in \mathbb{Q}$, $-1 = -\frac{1}{1} \in \mathbb{Q}$, by 1b, $-1(a) = -a$. By 6a, $-a \in \mathbb{Q}$. By claim 6a, since $a + b \in \mathbb{Q}$ and $-a \in \mathbb{Q}$, then $a + b - a \in \mathbb{Q}$. Since addition is commutative in the real numbers, thus $0 + b = a - a + b = a + b - a \in \mathbb{Q}$. By properties of the additive identity, $0 + b = b \in \mathbb{Q}$. This contradicts our assumption that b is irrational thus by contradiction $a + b \notin \mathbb{Q}$.

Suppose $a \neq 0$, assume the contrary that $ab \in \mathbb{Q}$. By definition of rational, $\exists p, q \in \mathbb{Z} \ni a = \frac{p}{q}, q \neq 0$. By the existence of a multiplicative identity for nonzero element in $\mathbb{R}$, $\exists \frac{1}{a} \in \mathbb{R} \ni \frac{1}{a} = \frac{1}{\frac{p}{q}} = \frac{q}{p}$. Since $q, p \in \mathbb{Z}$, then $\frac{1}{a} = \frac{q}{p} \in \mathbb{Q}$. By problem 6a, $\frac{1}{a}(ab) \in \mathbb{Q}$. By the fact that multiplication is associative, $\frac{1}{a}(ab) = (\frac{1}{a}a)b = 1b \in \mathbb{Q}$. By the properties of the multiplicative identity, $1b = b \in \mathbb{Q}$, which contradicts our assumption that b is irrational. Thus by contradiction, $ab \notin \mathbb{Q}$. $\square$

Problem 7. Show that there does not exist a rational number t such that $t^2 = 3$.

Proof. Assume the contrary that $\exists t \in \mathbb{Q} \ni t^2 = 3$. By the definition of rational numbers, $\exists p, q \in \mathbb{Z} \ni t = \frac{p}{q}, q \neq 0$. Assume $g = \gcd(p, q) \neq 1$, thus $p = ag$ and $q = bg$ where $a, b \in \mathbb{Z}$ and $\gcd(a, b) = 1$. Thus $t = \frac{p}{q} = \frac{ag}{bg} = \frac{a}{b}$. Thus without loss of generality, $\gcd(p, q) = 1$. Thus $(\frac{p}{q})^2 = \frac{p^2}{q^2} = 3$. By the existence

of a multiplicative inverse for any non zero element, $\exists \frac{1}{q^2} = q^2 \in \mathbb{R} \ni \frac{q^2}{q^2} = 1$. Thus $q^2 \frac{p^2}{q^2} = 1 * p^2 = 3q^2$. By the properties of the multiplicative identity, $1 * p^2 = p^2$. Thus $p^2 = 3q^2$. Thus p^2 is a factor of 3, thus 3 is also a factor of p . Since $\gcd(p, q) = 1$, 3 cannot divide q .

Thus $p = 3m$ for some $m \in \mathbb{Z}$. Thus $9m^2 = 3q^2$. There exists $\frac{1}{3} \in \mathbb{R} \ni 3 * \frac{1}{3} = 1$, thus $3m^2 = \frac{9}{3}m^2 = q^2$. Thus 3 is a factor of q^2 , thus 3 is also a factor q . Thus 3 divides q .

Thus 3 is a factor of q and is also not a factor of q , which leads to a contradiction. Thus since the statement that $\exists p, q \in \mathbb{Z} \ni q \neq 0, (\frac{p}{q})^2 = 3$ leads to a contradiction thus by contradiction, $\nexists t \in \mathbb{Q} \ni t^2 = 3$. Thus t must be irrational. $\square$